# Establishing Ground Truth: A Hybrid 3D-Printed & Synthetic Data Generation Framework For Molecular Imaging Assessment

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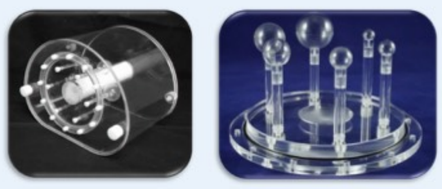
## BACKGROUND

**Molecular Imaging advances** — Dynamic imaging (4D), Theranostics, and AI are evolving rapidly

**Need for ground truth** — to validate new features, tools, protocols ***prior to*** clinical use

**Standardized Phantoms** (physical test objects) lack customizability and require tedious preparation highly prone to errors

**Limited validation diversity** — small clinical datasets and few controlled ground truths restrict robust assessment and confidence of new imaging tools



Typically used quality control (QC) phantoms

**Solution: A Complementary Framework**

**Tool A (Physical):** 3D Printed Phantoms to provide physical ground truth

**Tool B (Synthetic):** Simulation to provide analytical ground truth and extend assessment

## METHODS

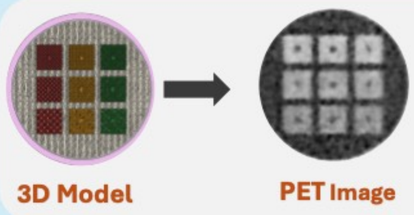
### 3D Printed "Analog" Ground Truth

**The Engineering Constraint:** Creating variable image contrast *without* mixing multiple liquid solutions / multiple compartments




**Porosity ($\phi$) as parameter to control dilution of radiotracer -> adjust image contrast**

Tissue Activity Concentration $(AC_{Bkg}) = \phi * AC_{Lesion}$




**Lesions appear more darker and tissue brighter**

- **Advantages of this Method:** Simple, Customizable, Reproducible, Multi-Contrast with single injection

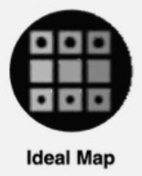
### Diffusion Decay Simulations with Synthetic Noise Addition

**The Computational Constraint:** Generating scalable, realistic datasets *without* the processing time of full Monte Carlo physics.

**Tracer Distribution**

$$\frac{\partial C_i}{\partial t} = \nabla.(\mathbf{D}.\nabla C_i) - (\lambda * C_i);\ \lambda = \frac{log\,2}{T_{1/2}}$$

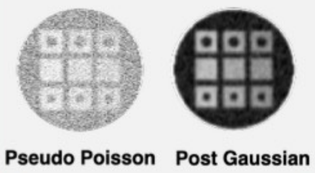
D: Diffusion Coefficient of Glucose
C: Concentration in mol/m3
$T_{1/2}$: Radiotracer Half life



Ideal Map

+

**Scanner Noise**

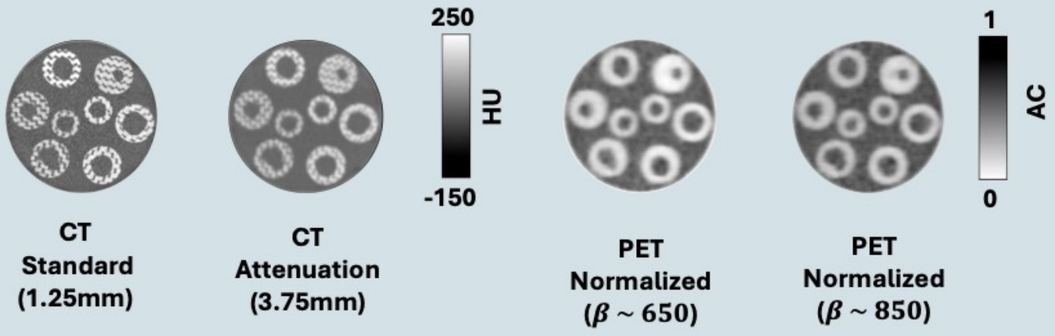



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**Simulated Image**

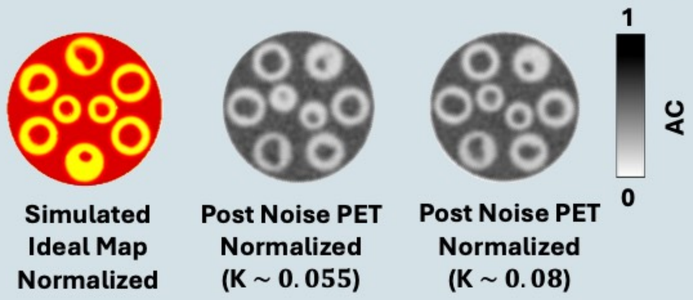


## RESULTS



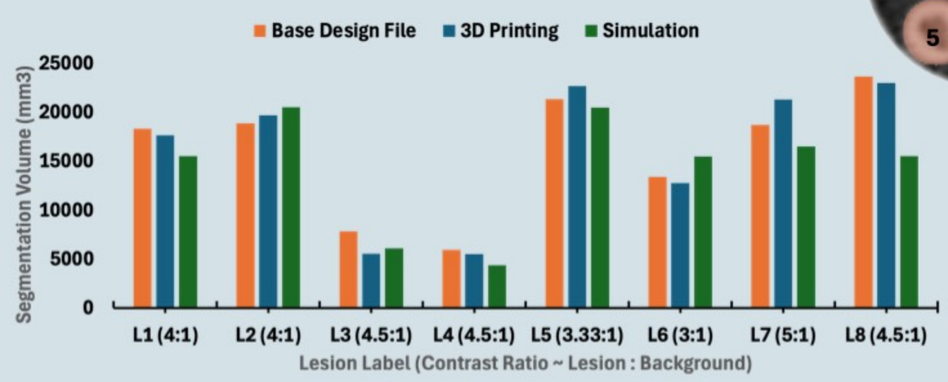

**Increase in beta (reconstruction smoothness) suppresses noise at low activity**

**! K is the noise calibration parameter used to match the mean variance ratio of simulation to acquisition**

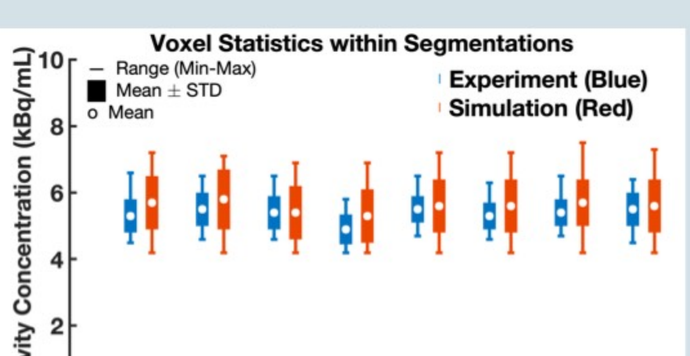



**Mean segmentation volume differences were 9% for Experiment and 16% for simulation data – Lower than typically used clinical margin of 20%**




For $\beta \sim 850$ & $K \sim 0.08$




- **Mean simulated values less than 10% variability**
- **Maximum values have 12.5% variability indicating slight overestimation**

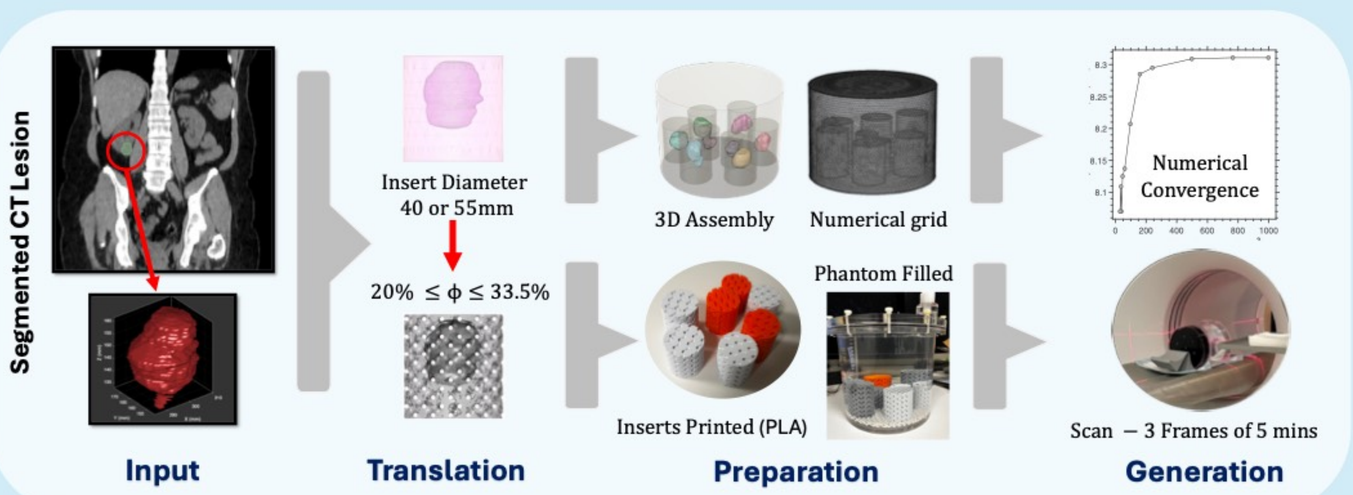
The synthetic model provides a reliable "Absolute Truth" for quantitative benchmarking

•**Strength:** Framework functions at complex textures and clinically realistic shapes
•**Limitation:** 3D Printing is limited to contrast ratios of **~5:1** due to printing limits
•**Solution: Simulation** component can model high-contrast target (>5:1)

### Anthropomorphic Lesion Evaluation – Pilot Study

**8 Patient Lesions extracted from The Cancer Archives (TCIA)**

**Purpose:** Bridging the gap between simple spheres and clinically realistic geometries




50MBq FDG $^{18}$F - 45 Minutes prior to scan

Clinical Acquisition & Reconstruction

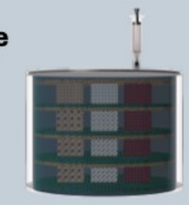
Metrics for Quantitative Inference

**65% Isolevel Segmentation**
**Calibrated to largest pososity (33.5%) with a 2.5% margin**

**Segmentation Volumes & Statistics**

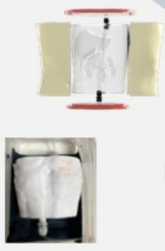
## PERSPECTIVES

**For QC - 36 Multi Contrast Targets with single injection - 6 times fewer experiments under same conditions**



**Customizable Solution:** Bridges the gap to clinically realistic ground truths

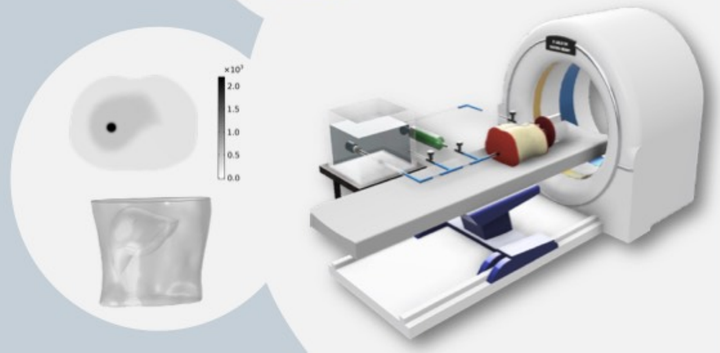
**Validated Reliability: <10% signal deviation** from physical ground truth



•**Dynamic Imaging Extension**: Time-activity variation for 4D imaging

**Future Work**

•**Expanded Lesion Database:** Integrating complex, **heterogeneous morphologies** (necrotic cores, irregular boundaries)



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References